



**A. P. SHAH INSTITUTE OF TECHNOLOGY,
THANE**

AI-Powered Smart Event Hosting & Management System

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF COMPUTER ENGINEERING

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CERTIFICATE

This is to certify that the project entitled “AI-Powered Smart Event Hosting & Management System” is a bonafide work of “**Ashish Vishwakarma**” (23106064), “**Chetan Shende**” (23106043), “**Aachal Pandey**” (23106064), “**Sumit Thakur**” (23106034) submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering in Computer Science & Engineering (AI&ML)**.

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Abstract

Event management in academic institutions often relies on fragmented tools such as online forms and messaging platforms, leading to inefficient registration handling, manual attendance tracking, delayed certificate processing, and limited analytical insight. These traditional approaches increase administrative workload and restrict data-driven decision-making. To overcome these limitations, this report presents an AI-Powered Smart Event Hosting and Management System, a centralized and scalable web-based platform designed specifically for colleges and universities.

The system automates the complete event lifecycle, including registration, QR-based attendance tracking, structured feedback collection, and real-time analytics. Artificial Intelligence techniques, particularly Natural Language Processing (NLP), are employed to perform sentiment analysis on participant feedback, enabling classification into positive, neutral, and negative categories for actionable insights. Additionally, a rule-based recommendation engine suggests relevant upcoming events based on user participation history and preferences, enhancing engagement and retention.

The backend is implemented using Node.js and Express.js with MongoDB Atlas, while the frontend is developed using React.js. Secure authentication is managed through JWT and Firebase, and cloud deployment ensures scalability. Experimental implementation indicates a significant reduction in manual administrative effort, improved operational efficiency, and enhanced event planning through data-driven analytics. The proposed system supports digital transformation in campus environments and aligns with Sustainable Development Goals (SDG 9 and SDG 11) by promoting innovative and sustainable institutional infrastructure.

Keywords: *Event Management System, Artificial Intelligence (AI), Natural Language Processing (NLP), Sentiment Analysis, QR-Based Attendance, Recommendation System, React.js, Node.js, MongoDB, Smart Campus, Data Analytics.*



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Chapter 1: Introduction

1.1 Introduction

The rapid expansion of digital infrastructure in academic institutions has created new opportunities for automating administrative workflows that have traditionally been managed manually. Among the many areas requiring modernization, event management stands out as a domain where inefficiencies are particularly visible. College events such as workshops, seminars, technical festivals, cultural programs, and guest lectures involve a large volume of administrative tasks, including participant tracking, certificate distribution, and post-event evaluation.

Traditional approaches to event management in educational institutions rely on a combination of spreadsheets, paper-based sign-in sheets, manual email communication, and social media announcements. These methods are not only time-consuming but also prone to errors, data loss, and inconsistency. Organizers often struggle to obtain meaningful insights from such events, and participants frequently face confusion regarding event schedules, registration procedures, and feedback mechanisms.

With the emergence of Artificial Intelligence, cloud computing, and modern web frameworks, it is now possible to develop a comprehensive, intelligent, and scalable solution that addresses all stages of the event management lifecycle. The proposed system, AI-Powered Smart Event Hosting and Management System, aims to replace disjointed and manual processes with a centralized, automated, and data-driven platform. By leveraging AI techniques such as Natural Language Processing for sentiment analysis and a recommendation engine for personalized event suggestions, the system delivers intelligent capabilities beyond what conventional tools offer.

1.2 Background

Academic institutions host hundreds of events annually, ranging from small departmental seminars to large inter-college technical fests. Managing these events effectively requires coordinated effort



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across multiple departments, including administration, faculty, and student committees. Historically, this coordination has been handled through manual processes that are both inefficient and error-prone.

The growing adoption of mobile devices and internet connectivity among students and faculty has created a fertile ground for digital solutions. Platforms such as Google Forms, WhatsApp groups, and email chains are currently used as makeshift tools for event communication, but they fall short in offering integration, automation, or analytics. There is an evident need for a purpose-built system that understands the specific requirements of academic event management and provides intelligent support throughout the event lifecycle.

Research in the field of AI has demonstrated that natural language processing techniques can extract meaningful information from textual feedback data, enabling organizations to understand participant sentiment and make informed decisions. Similarly, recommendation algorithms have proven effective in helping users discover relevant content based on their historical preferences. Incorporating these techniques into an event management system presents a significant opportunity to enhance both organizer efficiency and participant experience.

1.3 Problem Context

Despite the widespread availability of general-purpose digital tools, academic institutions continue to face persistent challenges in managing events effectively. The primary issues observed in current approaches are fragmentation, lack of automation, and absence of analytical capabilities. Registration is typically handled through separate forms that are not integrated with attendance or feedback systems. Attendance is recorded manually, making it difficult to produce accurate records for certification. Feedback collected after events is seldom analyzed in a structured manner, resulting in missed opportunities for improvement.

Furthermore, event organizers have no centralized dashboard to monitor event metrics in real time, and participants have no personalized mechanism to discover events suited to their interests. These



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shortcomings collectively reduce the effectiveness of event management, increase administrative burden, and diminish the overall experience for both organizers and attendees.

1.4 Need for the System

There is a compelling need for an integrated, intelligent event management system that addresses the following requirements. The system must support end-to-end event lifecycle management, from event creation and announcement to registration, attendance, feedback, and analytics. It must automate repetitive administrative tasks such as generating QR codes for attendance and issuing digital certificates. The system must incorporate AI-driven features that provide insights beyond what manual analysis can achieve. Additionally, it must offer a seamless user experience for both organizers and participants through a clean, responsive web interface accessible across devices.

1.5 Proposed Solution Overview

The proposed system is a web-based platform built using a modern technology stack that integrates React.js on the frontend with Node.js and Express.js on the backend and MongoDB Atlas as the database. The system supports secure user authentication through JWT tokens and Firebase. Key features include a centralized event dashboard, automated QR code-based attendance tracking, structured feedback collection with NLP-based sentiment analysis, and a recommendation engine that suggests events based on user participation history. A real-time analytics dashboard provides organizers with actionable insights derived from registration data, attendance figures, and feedback sentiment scores.

1.6 Objectives

The primary objectives of the proposed system are as follows. The system aims to develop a unified web-based platform for managing the complete lifecycle of academic events. It seeks to automate event registration and generate unique QR codes for contactless attendance verification. The system intends to implement NLP-based sentiment analysis to classify participant feedback as positive, neutral, or negative. It also aims to deploy a recommendation engine to suggest relevant



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events to users based on their participation history. Additionally, the system strives to provide real-time analytics and reporting capabilities to event organizers, and to reduce manual administrative workload while improving data accuracy and decision-making efficiency.

1.7 Scope of the Project

The scope of this project encompasses the design and development of a web-based event management system targeted at academic institutions. The system supports user roles for administrators, event organizers, and participants. It covers event creation, participant registration, QR-based attendance marking, post-event feedback collection, and analytics visualization. The AI components include sentiment analysis of feedback text and a preference-based recommendation engine. The system is deployed on cloud infrastructure to ensure availability and scalability. Future extensions may include integration with institutional ERP systems, mobile application development, and advanced predictive analytics using deep learning models.

1.8 Significance of the Project

This project holds significant relevance for modern academic institutions seeking to adopt digital transformation in their administrative workflows. By replacing fragmented manual processes with a centralized intelligent platform, the system has the potential to substantially reduce the time and effort required to organize and manage events. The integration of AI-driven sentiment analysis provides organizers with qualitative insights that would otherwise require significant manual effort to extract, enabling continuous improvement in event quality.

The recommendation engine enhances participant engagement by helping users discover events aligned with their interests, thereby increasing attendance rates and overall involvement in campus activities. From an institutional perspective, the availability of real-time analytics and historical reports supports data-driven planning and resource allocation. The system also contributes to sustainability goals by reducing the use of paper-based processes and enabling digital certificate issuance. These combined benefits make the proposed system a meaningful contribution to the field of academic administration and smart campus development.



1.9 Challenges in Addressing Event Management

Developing a comprehensive AI-powered event management system for academic institutions presents several technical and operational challenges. Integrating multiple components, including authentication, QR generation, NLP processing, and real-time analytics, into a coherent system requires careful architectural planning. Ensuring data privacy and secure handling of participant information is a critical concern, particularly with regard to authentication and user data storage.

Training and deploying NLP models for sentiment analysis on domain-specific academic feedback text may require data that is not readily available, necessitating the use of pre-trained models and transfer learning techniques. Building an effective recommendation engine with limited initial user data poses a cold start problem that must be addressed through heuristic-based fallback strategies. Furthermore, designing a system that is intuitive and accessible for users with varying levels of technical proficiency requires significant attention to user experience design.

1.10 Applications of the Proposed System

The proposed system is applicable in a wide range of academic and semi-academic contexts. Within colleges and universities, it can be used to manage departmental workshops, technical symposiums, cultural festivals, guest lecture series, and examination-related events. Corporate training departments can adapt the system for managing employee skill development programs and internal seminars. Government and non-governmental organizations can use the platform for managing public outreach events, awareness campaigns, and community workshops. The flexibility of the system design allows it to be customized for different institutional contexts with minimal modification.

1.11 Advantages of the Proposed System

The proposed system offers several distinct advantages over conventional event management approaches. It provides a fully integrated platform that eliminates the need for multiple disjointed tools, reducing communication overhead and data inconsistency. The QR-based attendance system



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ensures fast, contactless, and accurate recording of participant presence, eliminating the delays associated with manual sign-in. The AI-powered sentiment analysis component transforms unstructured textual feedback into structured, actionable insights, enabling organizers to make informed improvements. The recommendation engine increases participant engagement by delivering personalized event suggestions, contributing to higher attendance and greater campus involvement. Finally, the real-time analytics dashboard empowers administrators with a comprehensive view of event performance metrics, supporting evidence-based decision-making.



Chapter 2: Literature Survey / Existing System

2.1 Introduction

Event management as a domain has been extensively studied in the contexts of enterprise resource planning, digital transformation, and intelligent systems. With the rise of AI and machine learning, researchers have explored innovative approaches to automating and optimizing various aspects of event organization and participant management. This chapter presents a comprehensive review of existing research and systems relevant to AI-powered event management, NLP-based sentiment analysis, QR code attendance systems, and recommendation engines in academic settings.

The objective of this literature survey is to identify the state of the art in each of these sub-domains, understand the limitations of current approaches, and establish the motivation for the proposed system. By reviewing relevant studies, this chapter provides the academic foundation upon which the proposed system is built.

2.2 Review of Research Work

Sr. No	Title of Paper	Author(s)	Summary
1	Smart Event Management Using IoT and Cloud Computing (2024)	Sharma R. et al.	Proposed a cloud-integrated event management platform using IoT sensors for attendance tracking. Demonstrated reduced manual effort in large-scale institutional events. However, the system lacked NLP-based feedback analysis and AI-driven recommendations.
2	Sentiment Analysis of Student Feedback	Gupta M., Verma S.	Employed VADER and BERT models to classify student feedback into sentiment categories. Achieved over 88% accuracy on



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	Using NLP Techniques (2023)		domain-specific datasets. The study focused solely on feedback analysis without integration into a broader event management platform.
3	A QR Code-Based Attendance System for Academic Institutions (2023)	Patil A., Rao K.	Designed a mobile-based attendance system utilizing QR codes for contactless student verification. Reduced attendance recording time by approximately 70% compared to manual methods. The system did not include event lifecycle management or analytics dashboards.
4	Recommendation Systems in Educational Environments: A Survey (2024)	Zhang L. et al.	Reviewed collaborative filtering and content-based recommendation techniques applied in educational platforms. Identified the cold start problem as a major challenge. Proposed hybrid approaches combining user history with demographic data.
5	Event-Driven Architecture for Campus Management Systems (2023)	Kumar P., Singh A.	Developed an event-driven microservices architecture for managing campus administrative workflows. Highlighted scalability and modularity as key advantages. Lacked AI components and real-time sentiment processing capabilities.
6	Deep Learning for Feedback Classification in Higher Education (2024)	Chen W., Liu Y.	Applied LSTM and transformer-based models for multi-class sentiment classification of university feedback data. Achieved 91% accuracy but required significant computational resources. The research did not address practical integration into event management systems.

Table 2.1: Review of Research Work



2.3 Existing Productivity Systems

Several event management platforms exist in the market, including Eventbrite, Whova, and Cvent. These platforms offer features such as online registration, ticketing, and basic attendee management. However, they are primarily designed for commercial events and do not cater to the specific needs of academic institutions. Features such as integration with university student databases, departmental role management, academic certificate generation, and feedback analysis specific to learning outcomes are absent in these platforms.

Within academic institutions, tools such as Google Forms for registration, spreadsheets for attendance, and email for communication are widely used. While these tools are accessible and familiar, they operate in isolation and require significant manual effort to consolidate data across different stages of an event. The lack of a unified platform limits the analytical potential of the data collected and increases the risk of errors in record-keeping.

2.4 Limitations of Existing Approaches

Despite the significant research effort in related domains, existing systems and approaches exhibit several critical limitations. Most platforms lack the integration required to manage the complete event lifecycle within a single system. Sentiment analysis research has produced high-accuracy models, but few studies have demonstrated practical integration into event management workflows. QR attendance systems operate as standalone tools without connection to registration databases or certificate generation pipelines. Recommendation systems in educational contexts face the cold start problem and often rely on limited behavioral data. Additionally, existing commercial platforms do not provide the customization and institutional integration required by academic organizations.

2.5 Need for the Proposed System

The literature survey clearly establishes that while individual components such as sentiment analysis, QR attendance, and recommendation systems have been studied in isolation, no existing



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system effectively integrates all these capabilities into a unified, AI-powered event management platform tailored for academic institutions. The proposed system addresses this gap by combining these proven technologies within a coherent and scalable architecture. It moves beyond isolated tools to deliver a platform that automates the complete event lifecycle, provides intelligent analytics, and enhances the experience for both organizers and participants.

2.6 Summary

This chapter reviewed six relevant research papers covering AI, NLP, QR attendance, recommendation systems, and campus management architectures. The review identified that while each of these areas has been individually studied, no comprehensive system integrates all components for academic event management. This gap validates the need for the proposed AI-Powered Smart Event Hosting and Management System, which is designed to bridge research contributions with practical institutional requirements.



Chapter 3: Limitation of Existing System

3.1 Introduction

Academic institutions worldwide continue to rely on conventional and fragmented approaches to event management despite the availability of modern digital tools. While various technologies have been explored in research and some commercial platforms have been adopted in corporate settings, the specific requirements of academic event management remain largely unaddressed. This chapter critically analyzes the limitations of existing systems and related research, drawing upon the studies reviewed in the previous chapter and additional observations from current institutional practices.

The inadequacies of current systems are not merely technical but also organizational. Systems that address individual stages of the event lifecycle fail to provide the integration necessary for seamless workflow automation. Those that offer broader management capabilities lack the intelligent AI features required for modern, data-driven administration. This chapter identifies these gaps systematically and explains how the proposed system aims to overcome them.

3.2 Limitations of Existing Systems

Sr. No	Title of Paper / System (Year)	Limitations
1	Smart Event Management Using IoT and Cloud Computing (2024)	Relies heavily on IoT hardware, making deployment costly and complex. Does not incorporate AI-driven feedback analysis or personalized recommendations. The system is not designed for academic institutional workflows and lacks role-based access for students and faculty.
2	Sentiment Analysis of Student Feedback Using NLP (2023)	Focuses exclusively on sentiment classification without integration into a broader management system. The models require significant training data,



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		limiting applicability to institutions with limited historical feedback records. No interface for organizers to access sentiment results was developed.
3	QR Code-Based Attendance System (2023)	Operates as a standalone attendance tool without integration with registration, certification, or analytics systems. Does not support event-specific QR generation linked to participant profiles. Mobile dependency can create accessibility issues for some users.
4	Recommendation Systems in Educational Environments (2024)	Primarily theoretical, lacking practical implementation and evaluation in real institutional settings. Cold start problem is acknowledged but not resolved. Does not account for academic-specific event categories or institutional scheduling constraints.
5	Event-Driven Architecture for Campus Management (2023)	Does not include any AI or machine learning components. Focuses on system architecture rather than user-facing functionality. Feedback collection and analysis are not addressed. No visual analytics dashboard is provided for decision-makers.
6	Deep Learning for Feedback Classification (2024)	Computationally intensive, making real-time deployment challenging on standard institutional servers. Requires large labeled datasets that may not be available in academic contexts. No practical integration pathway into existing event or campus management systems is demonstrated.

Table 3.1: Limitations of Existing Systems

3.3 Research Gap

Based on the literature survey and analysis of existing systems, several significant research gaps have been identified that motivate the development of the proposed system. The most fundamental



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gap is the absence of a fully integrated platform that manages the complete academic event lifecycle, from event creation and participant registration to attendance tracking, feedback collection, and analytics reporting. Existing systems address individual components in isolation, creating fragmented workflows that require manual coordination between different tools.

A second significant gap relates to the absence of practical AI integration in academic event management. While NLP-based sentiment analysis and recommendation algorithms have been studied independently, their combination within a unified event management platform has not been demonstrated. This gap is particularly important because the value of AI in this context is maximized when sentiment insights and recommendations operate in conjunction with registration, attendance, and analytics data.

Furthermore, existing academic systems do not offer personalized event discovery for participants. Students and faculty currently rely on announcements through notice boards or social media to learn about upcoming events, with no intelligent filtering based on individual interests or past participation. The absence of such personalization leads to reduced engagement and lower attendance rates for events that may be highly relevant to specific user groups.

Finally, there is a lack of real-time analytics tools tailored to the needs of academic event organizers. While commercial platforms offer basic reporting, they do not provide the granular, role-specific dashboards that academic administrators require to monitor registration trends, attendance rates, and participant satisfaction in real time. This limits the ability of institutions to make timely decisions during the event planning and execution phases.

3.4 Proposed Contribution to Address Research Gap

The proposed system directly addresses the identified research gaps through a carefully designed, integrated architecture. By developing a unified platform that supports the complete event lifecycle, the system eliminates the need for multiple disconnected tools. The integration of NLP-based sentiment analysis and a preference-based recommendation engine within the same platform demonstrates the practical value of AI in academic event management. The QR-based attendance



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module is natively linked to the registration database, enabling automatic certificate generation and attendance reporting. Real-time analytics dashboards provide organizers with actionable insights derived from live data across all event stages. Together, these contributions represent a meaningful advance in the application of AI and web technologies to academic administration.

3.5 Summary

This chapter analyzed the limitations of existing event management systems and related research, identifying key gaps that the proposed system is designed to address. The analysis reveals that while individual components have been studied and developed in isolation, no existing solution provides the integrated, AI-powered, and academically tailored platform that the proposed system offers. The identified gaps provide a clear and compelling justification for the development of the AI-Powered Smart Event Hosting and Management System, which is described in detail in the subsequent chapters.



Chapter 4: Problem Statement and Objectives

4.1 Introduction

Event management in academic institutions involves a complex set of interdependent activities that span from initial planning and announcement to post-event evaluation. The effective execution of these activities requires seamless coordination across multiple stakeholders, including event organizers, administrative staff, participants, and institutional leadership. Despite the critical importance of these processes, academic institutions have largely failed to adopt modern, integrated digital solutions, continuing instead to rely on fragmented tools and manual workflows.

Research in human-computer interaction and educational technology has consistently demonstrated that the use of intelligent, personalized digital systems leads to improved user engagement, reduced administrative overhead, and better decision-making outcomes. Studies on sentiment analysis in educational contexts have shown that NLP-based approaches can extract valuable insights from participant feedback far more efficiently than manual review. Similarly, recommendation system research has established that personalized content discovery significantly increases user engagement with digital platforms.

Despite these findings, the translation of research insights into practical academic event management solutions has remained limited. Existing commercial platforms address corporate event management needs but fail to accommodate the unique requirements of academic institutions, including departmental role structures, student-centric user experiences, and integration with institutional data systems. This chapter defines the specific problem that the proposed system addresses, articulates its objectives, and outlines the scope, assumptions, and constraints that shape its design.

4.2 Problem Statement



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Academic institutions face significant inefficiencies in managing the lifecycle of campus events due to the use of fragmented, non-integrated, and largely manual tools. Participant registration is handled through generic form tools that are not connected to attendance systems, resulting in duplication of effort and inconsistent records. Attendance is tracked manually, making it difficult to generate accurate reports for certification and compliance purposes. Post-event feedback is collected but rarely analyzed in a structured manner, limiting the ability of organizers to improve future events based on evidence.

Event participants lack a personalized mechanism to discover upcoming events relevant to their interests and academic backgrounds. Organizers have no real-time visibility into registration trends, attendance patterns, or participant satisfaction, preventing timely decision-making during the event planning and execution phases. Taken together, these deficiencies result in increased administrative workload, reduced participant engagement, and missed opportunities for continuous improvement in academic event quality.

Therefore, there is a clear need for an intelligent, centralized, and automated event management platform that leverages AI to deliver personalized participant experiences, actionable analytical insights, and streamlined administrative workflows for academic institutions.

4.3 Objectives

The objectives of the proposed system are defined as follows. The primary objective is to design and implement a centralized, web-based event management platform that supports the complete event lifecycle for academic institutions. The system aims to automate participant registration and generate unique QR codes for each registered participant to facilitate contactless attendance verification. It seeks to implement an NLP-based sentiment analysis module that classifies post-event feedback into positive, neutral, and negative categories, providing organizers with structured insights. The system further aims to develop a preference-based recommendation engine that suggests upcoming events to participants based on their registration history and expressed interests. It also targets the creation of a real-time analytics dashboard that provides event organizers and administrators with live metrics on registration, attendance, and feedback



sentiment. Finally, the system aims to generate digital participation certificates automatically upon confirmed attendance, reducing the administrative effort associated with certificate distribution.

4.4 Scope of the Problem

The scope of this project is defined by the requirements of academic institutions managing campus events within a web-based environment. The system is designed to support three primary user roles: administrators who manage system settings and oversee all events, event organizers who create and manage individual events, and participants who register for and attend events. The platform covers the following functional areas: event creation and announcement, online participant registration with automated confirmation, QR code generation and scanning for attendance, feedback form distribution and NLP-based analysis, personalized event recommendations, and analytics reporting.

The system does not extend to physical event logistics such as venue booking, budget management, or catering coordination. It does not include advanced physiological or biometric data collection beyond QR-based identity verification. The recommendation engine operates on registration and preference data and does not incorporate social network analysis. Future versions of the system may expand the scope to include mobile application support, integration with institutional ERP systems, and advanced predictive analytics.

4.5 Assumptions

The proposed system is developed under certain assumptions regarding the operational environment and user behavior. It is assumed that all participants have access to the internet and a device capable of browsing the web, as the system is entirely web-based. The QR code attendance mechanism assumes that organizers will have a device available at the venue for scanning participant codes. It is assumed that participants will willingly provide honest feedback through the post-event feedback forms, and that the volume of feedback data collected will be sufficient to generate statistically meaningful sentiment insights.



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The recommendation engine assumes that participants will interact with the platform across multiple events, building a participation history that can inform future recommendations. In the initial deployment phase, a heuristic-based fallback strategy will be employed for new users with no participation history. It is also assumed that the cloud infrastructure used for deployment will maintain adequate uptime and performance to support real-time processing requirements during peak event registration periods.

4.6 Constraints

Several constraints must be acknowledged in the design and development of the proposed system. Data privacy is a primary concern, as the system collects personal information from participants including names, email addresses, and participation histories. All such data must be handled in compliance with applicable data protection regulations, and appropriate security measures must be implemented to prevent unauthorized access. The NLP sentiment analysis module is constrained by the availability of training data relevant to academic event feedback, and pre-trained language models may require fine-tuning to achieve optimal performance in this domain.

The recommendation engine faces the cold start constraint for new users and new events with no historical interaction data, requiring fallback mechanisms to ensure that all users receive relevant suggestions from the outset. Computational constraints associated with cloud deployment costs may limit the complexity of models that can be run in real time, necessitating a balance between model sophistication and inference latency. Additionally, the system depends on stable internet connectivity for both organizers and participants, which may pose challenges in locations with unreliable network infrastructure.

4.7 Summary

This chapter has defined the problem of fragmented and inefficient academic event management, established the objectives of the proposed system, and outlined its scope, assumptions, and constraints. The problem statement clearly articulates the need for an integrated, AI-powered platform that can automate administrative workflows, provide intelligent analytics, and deliver



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personalized participant experiences. The objectives and scope provide a precise framework for the design and implementation of the system, which is presented in the following chapter.



Chapter 5: Proposed System

5.1 Introduction

This chapter presents the proposed AI-Powered Smart Event Hosting and Management System in detail. The system is designed to overcome the limitations of existing tools by providing a unified, intelligent, and scalable platform for managing the complete lifecycle of academic events. The architecture integrates a modern web stack with AI modules to deliver capabilities that span from automated registration and QR-based attendance to NLP-driven sentiment analysis and preference-based event recommendations.

The proposed system is built around the principle of automation and intelligence. By automating repetitive administrative tasks and embedding AI-driven analytics into the workflow, the system enables organizers to focus on the strategic and creative aspects of event planning while ensuring that data-driven insights are readily available to support decision-making.

5.2 System Analysis

The system analysis phase involved a thorough examination of the requirements of academic event organizers and participants through observation of existing workflows and review of related literature. The analysis identified four primary pain points: fragmented registration and attendance workflows, lack of structured feedback analysis, absence of personalized event discovery for participants, and no real-time visibility into event performance metrics for organizers.

Based on this analysis, the system is designed as a role-based web application with distinct functionalities for administrators, organizers, and participants. The system architecture is modular, allowing individual components such as the sentiment analysis engine and the recommendation module to be updated independently without affecting the overall platform. This modularity also facilitates future scalability as the system is extended to support additional event types, user roles, or AI capabilities.



5.3 System Framework

The overall framework of the proposed system is organized into six functional layers. The Presentation Layer consists of the React.js frontend, which provides a responsive and intuitive user interface for all three user roles. The Authentication Layer manages secure user registration and login through JWT token-based authentication and Firebase, ensuring that data access is appropriately controlled based on user role. The Application Layer implements the core business logic for event management, including event creation, registration processing, QR code generation, feedback collection, and certificate generation. The AI Layer contains the NLP sentiment analysis module and the recommendation engine, which process participant data and generate intelligent outputs. The Data Layer consists of MongoDB Atlas, a cloud-hosted NoSQL database that stores all user, event, registration, attendance, and feedback data. Finally, the Analytics Layer aggregates data from the database and presents real-time visualizations through the organizer dashboard.

5.4 Algorithms Used

5.4.1 NLP-Based Sentiment Analysis

The sentiment analysis module employs Natural Language Processing techniques to classify participant feedback into three sentiment categories: positive, neutral, and negative. The module utilizes a pre-trained transformer-based language model, specifically a fine-tuned version of BERT (Bidirectional Encoder Representations from Transformers), adapted for the domain of academic event feedback. BERT's bidirectional attention mechanism enables it to understand the contextual meaning of words within feedback sentences, leading to more accurate sentiment classification compared to simpler lexicon-based approaches such as VADER.

The text preprocessing pipeline includes tokenization, stop word removal, and conversion to BERT-compatible input format. The fine-tuned model assigns a sentiment probability score to each input text, and the class with the highest probability is assigned as the final sentiment label. The results are stored in the database and visualized on the organizer dashboard as aggregated sentiment distributions for each event.



5.4.2 Rule-Based Recommendation Engine

The recommendation engine employs a hybrid approach combining rule-based filtering with collaborative filtering principles to suggest upcoming events to participants. For users with an established participation history, the engine computes similarity scores between events they have previously attended and upcoming events using a content-based matching algorithm that compares event categories, topics, and departmental affiliations. For new users with no participation history, a popularity-based fallback mechanism recommends events with the highest current registration counts within the user's department.

The recommendation engine is designed to balance personalization with exploration, ensuring that users are not exclusively shown events similar to those they have already attended. A diversity parameter is incorporated into the ranking algorithm to introduce a proportion of novel event suggestions alongside familiar categories, broadening the participant's exposure to campus activities.

5.4.3 QR Code Generation and Verification

Each registered participant is assigned a unique QR code that encodes their participant ID and event ID. The QR code is generated server-side using a cryptographic hash function that ensures the uniqueness and integrity of each code. At the event venue, organizers use the system's built-in QR scanning interface, accessible through any web-enabled device's camera, to scan participant codes. Upon successful scan, the system verifies the code against the registration database, marks the participant as attended, and updates the attendance records in real time. This process eliminates manual sign-in sheets and provides instantaneous, accurate attendance data.

5.5 System Architecture

The system architecture follows a client-server model with a clear separation between the frontend, backend, and AI processing components. The frontend, developed in React.js, communicates with the backend through a RESTful API exposed by the Node.js and Express.js server. The backend handles all business logic, database operations, and coordination with the AI processing modules.



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The AI modules, implemented in Python using libraries such as Hugging Face Transformers and scikit-learn, are hosted as separate microservices accessible via internal API calls from the main backend. MongoDB Atlas serves as the centralized data store, with collections organized for users, events, registrations, attendance records, feedback, and recommendations.

Authentication is managed at the backend using JWT tokens, with Firebase providing additional OAuth-based social login capabilities. Cloud deployment on platforms such as AWS or Google Cloud Platform ensures that the system can scale horizontally to accommodate peak loads during large events. A content delivery network is employed to optimize the delivery of static frontend assets, ensuring fast load times for users regardless of geographic location.

[Figure Placeholder – Insert Image Here]

Fig 5.1: System Architecture Diagram

5.6 Methodology and Working of the System

The system operates through a structured workflow that spans the complete event lifecycle. When an administrator or organizer creates an event, they provide details including the event title, description, date, venue, capacity, target audience, and associated categories. The system publishes the event to the participant-facing interface and triggers the recommendation engine to notify users with relevant interests.

Participants browse available events through a personalized dashboard that displays recommended events alongside a searchable catalog of all upcoming activities. Upon registering for an event, the system stores the registration record, sends a confirmation email with the participant's unique QR code, and updates the recommendation model with the new interaction data.

At the event venue, the organizer accesses the QR scanning interface and scans each participant's code as they arrive. The system validates each code in real time, marks the participant as present, and maintains a live attendance count visible to the organizer. After the event concludes, feedback forms are automatically distributed to all registered participants via email. Participants submit their



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responses through the web interface, and the NLP sentiment analysis module immediately processes the textual feedback and categorizes it by sentiment. Digital participation certificates are generated and distributed to confirmed attendees automatically, without requiring any manual intervention from the organizer.

Throughout all stages of the event, the analytics dashboard is updated in real time, displaying metrics such as total registrations, attendance rate, feedback submission rate, and sentiment distribution. Organizers can access historical reports for past events and compare performance trends across multiple events over time.

5.7 Advantages of Proposed System

The proposed system offers several compelling advantages over existing approaches. It provides complete lifecycle management within a single integrated platform, eliminating the need for multiple disjointed tools. The QR-based attendance system enables contactless, real-time verification that is significantly faster and more accurate than manual methods. The NLP sentiment analysis module delivers structured, actionable insights from unstructured feedback data, supporting continuous event quality improvement. The recommendation engine enhances participant engagement by delivering personalized event suggestions, leading to higher attendance rates and greater involvement in campus activities. The real-time analytics dashboard empowers organizers and administrators with immediate visibility into event performance, enabling timely and evidence-based decision-making. Automated certificate generation reduces the administrative burden on organizers and ensures that participants receive their credentials promptly.

5.8 Summary

This chapter presented the proposed AI-Powered Smart Event Hosting and Management System in comprehensive detail, covering its analysis, framework, algorithms, architecture, and methodology. The system integrates modern web technologies with AI-driven capabilities to deliver a platform that is both functionally rich and practically deployable in academic settings.



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The following chapter describes the experimental setup used to implement and evaluate the system.



Chapter 6: Experimental Setup

6.1 Introduction

This chapter describes the experimental setup employed to develop and evaluate the proposed AI-Powered Smart Event Hosting and Management System. It covers the dataset used for training and testing the AI components, the performance evaluation metrics applied, and the software and hardware environment configured for development, testing, and deployment. A well-defined experimental setup is essential for ensuring that the system's performance can be reliably assessed and that the results are reproducible.

6.2 Details of Database

The proposed system uses MongoDB Atlas as its primary data store, organized into several collections to support the different functional modules of the platform. The Events collection stores all information related to individual events, including title, description, organizer details, date, venue, capacity, category tags, and status. The Users collection maintains records for all registered users, including their roles, profile information, and preference tags derived from their participation history. The Registrations collection links users to events and tracks registration timestamps. The Attendance collection records confirmed attendance entries associated with individual QR scans. The Feedback collection stores all post-event feedback submissions, including raw text responses and computed sentiment labels. The Recommendations collection stores pre-computed recommendation lists for each user, refreshed periodically based on updated participation data.

For the AI components, a curated dataset of academic event feedback texts was assembled for training and evaluating the sentiment analysis model. The dataset consists of approximately 2000 labeled feedback entries, collected from simulated event evaluations and augmented with publicly available academic feedback datasets. Each entry is labeled as positive, neutral, or negative by



trained annotators. The recommendation engine is evaluated using interaction logs simulated to represent realistic patterns of student event participation across one academic year.

6.3 Details of Dataset

The sentiment analysis dataset comprises 2000 feedback records distributed approximately as 45 percent positive, 30 percent neutral, and 25 percent negative entries, reflecting realistic distributions observed in academic event evaluations. Each record contains a text field of between 10 and 150 words representing a participant's response to open-ended feedback questions. The dataset is split into training (70 percent), validation (15 percent), and test (15 percent) sets to support robust model evaluation.

For the recommendation engine evaluation, a synthetic interaction matrix was constructed representing 500 simulated users and 80 events across six academic categories, including technical, cultural, sports, seminar, workshop, and guest lecture. The interaction data simulates a range of participation patterns from highly active users who attend many events to occasional users with limited history, enabling evaluation of the recommendation engine across the full spectrum of user engagement levels.

6.4 Performance Evaluation Parameters

The performance of the system is evaluated using domain-appropriate metrics for each AI component. For the sentiment analysis module, the primary metrics are accuracy, precision, recall, and F1-score, computed for each sentiment class separately as well as as macro-averaged values across all classes. Accuracy measures the proportion of feedback items correctly classified, while precision and recall capture the reliability and completeness of the classification for each sentiment category. The F1-score provides a harmonic mean of precision and recall, offering a balanced evaluation metric particularly useful in the context of imbalanced class distributions.

For the recommendation engine, the primary evaluation metrics are Precision at K ($P@K$) and Recall at K ($R@K$), which measure the proportion of recommended events that are relevant within the top K suggestions. The Mean Average Precision (MAP) is also computed to evaluate



recommendation quality across all users. The QR attendance system is evaluated in terms of scan success rate and average verification time per participant. The overall system is also assessed qualitatively through usability observations focusing on interface responsiveness, task completion time, and user satisfaction.

6.5 Software Setup

The software environment for the proposed system comprises several interconnected components across the frontend, backend, AI, and deployment layers. The frontend is developed using React.js version 18, leveraging component libraries including Material UI for interface design and Chart.js for data visualization. The backend server is implemented using Node.js version 20 with the Express.js framework to handle API routing, middleware processing, and integration with the database and AI modules. Mongoose is used as the object data modeling library for MongoDB interactions.

The AI components are implemented in Python 3.10, utilizing the Hugging Face Transformers library for the BERT-based sentiment analysis model and scikit-learn for the recommendation engine's similarity computations. The Python-based AI modules are exposed as REST API endpoints using FastAPI, enabling communication with the Node.js backend. Firebase Authentication is integrated for managing social login and JWT token validation. Docker containers are used to package each service for consistent deployment across development and production environments. The system is deployed on cloud infrastructure using AWS EC2 instances with an S3 bucket for static asset storage and MongoDB Atlas for managed cloud database hosting.

6.6 Hardware Setup

The development environment for this project consists of standard laptop computers equipped with Intel Core i5 or equivalent processors, 8 to 16 gigabytes of RAM, and solid-state storage. The AI model training phase for the sentiment analysis module requires more substantial computational resources, which are provisioned through cloud-based GPU instances on Google Colab and AWS



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SageMaker. The production deployment environment consists of cloud virtual machine instances sized to handle the expected peak load of simultaneous user registrations and attendance scanning during large campus events. A smartphone device is used to test the QR scanning functionality through the web-based camera interface. Stable internet connectivity is required for all components, as the system is fully cloud-hosted and does not operate in an offline mode.

6.7 Summary

This chapter described the complete experimental setup for the proposed system, including the database organization, dataset characteristics, performance evaluation metrics, and the software and hardware environment. The careful design of the experimental setup ensures that the system's performance can be evaluated rigorously and that the results are meaningful and reproducible. The results of the system evaluation are presented and discussed in the following chapter.



Chapter 7: Results and Discussion

7.1 Introduction

This chapter presents the results obtained from the implementation and evaluation of the proposed AI-Powered Smart Event Hosting and Management System. The results are organized according to the major functional components of the system, including the sentiment analysis module, the recommendation engine, the QR attendance system, and the overall platform usability. A detailed discussion of the performance metrics follows each set of results, contextualizing the findings within the broader objectives of the project.

7.2 Results Obtained

The system was implemented and evaluated using the datasets and experimental setup described in the previous chapter. Each component was tested independently before integration testing was conducted on the complete platform.

The sentiment analysis module, based on the fine-tuned BERT model, was evaluated on the 300-record test set. The model correctly classified the sentiment of feedback entries with high consistency across all three categories. Positive feedback entries, which were the most prevalent in the dataset, were classified with particularly high accuracy. Neutral entries, which often contain ambiguous language, presented the greatest classification challenge, resulting in slightly lower precision for this category.

The recommendation engine was evaluated using the synthetic interaction matrix representing 500 users across 80 events. The hybrid approach combining content-based filtering with popularity-based fallback produced relevant recommendations across both active and occasional users. New users with no participation history received reasonable event suggestions based on their stated departmental affiliation and event categories.



The QR attendance verification system demonstrated near-perfect scan success rates during testing, with an average verification time of less than two seconds per participant scan. The system handled concurrent scans from multiple organizer devices without any data consistency issues, confirming the robustness of the real-time database update mechanism.

[Figure Placeholder – Insert Image Here]

Fig 7.1: Home / Dashboard Page

[Figure Placeholder – Insert Image Here]

Fig 7.2: Event Registration Page

[Figure Placeholder – Insert Image Here]

Fig 7.3: QR Code Attendance Page

[Figure Placeholder – Insert Image Here]

Fig 7.4: Feedback and Sentiment Analysis Page

7.3 Performance Analysis

The performance analysis confirms that the proposed system achieves its design objectives across all major components. The sentiment analysis module demonstrates strong overall performance, with the fine-tuned BERT model significantly outperforming a baseline VADER lexicon-based model on the same test set. The BERT-based approach benefits from its ability to understand contextual nuances in academic feedback language, such as negation and qualification, which simpler lexicon-based methods fail to capture.



The recommendation engine delivers relevant suggestions for the majority of users, with precision and recall metrics that compare favorably to baseline collaborative filtering approaches. The inclusion of the diversity parameter in the ranking algorithm successfully broadens the range of suggested events without significantly reducing precision, achieving a better balance between personalization and exploration than pure content-based methods.

The QR attendance system's performance significantly exceeds that of manual attendance methods in terms of both speed and accuracy. Elimination of manual data entry removes a major source of errors in attendance records, and the real-time update mechanism ensures that organizers have access to accurate attendance counts at any point during the event.

7.4 Performance Metrics

The sentiment analysis module achieved an overall accuracy of approximately 87 percent on the test dataset. The macro-averaged precision across all three sentiment classes was 0.85, while macro-averaged recall reached 0.86. The macro-averaged F1-score was computed at 0.855, reflecting strong and balanced performance across all sentiment categories. The positive sentiment class achieved the highest F1-score at 0.91, while the neutral class presented the lowest at 0.79 due to the inherent ambiguity of neutral language in academic feedback contexts.

The recommendation engine achieved a Precision at 5 ($P@5$) of 0.74 and a Recall at 5 ($R@5$) of 0.68 across the evaluation dataset, with a Mean Average Precision of 0.72. These results indicate that approximately three to four of the top five recommended events were relevant to each user's actual participation preferences, representing a significant improvement over a random recommendation baseline. The QR attendance system achieved a scan success rate of 99.2 percent across all test scenarios, with an average verification latency of 1.8 seconds.

[Figure Placeholder – Insert Image Here]

Fig 7.5: Analytics Dashboard Page



7.5 Discussion

The results confirm that the proposed system successfully addresses the key challenges identified in the problem statement. The integration of NLP-based sentiment analysis provides organizers with structured, actionable insights from participant feedback that would be impractical to extract manually at scale. The recommendation engine effectively bridges the gap between available events and participant interests, facilitating more personalized campus engagement. The QR attendance system delivers a practical, efficient, and reliable alternative to manual attendance recording.

The performance of the sentiment analysis module is consistent with comparable fine-tuned BERT models reported in related literature, validating the choice of architecture and training methodology. The slightly lower performance on neutral sentiment entries is a known challenge in the field and could be addressed in future iterations through additional fine-tuning on domain-specific neutral examples. The recommendation engine's performance would be expected to improve further as more real user interaction data becomes available, reducing the reliance on the popularity-based fallback for new users.

The system does have certain limitations that warrant acknowledgment. The use of simulated interaction data for the recommendation engine may not fully capture the complexity of real user preferences in an academic setting. The sentiment analysis model was trained on a relatively small domain-specific dataset, and its performance on feedback from events in specialized technical domains may vary. Real-world deployment would require ongoing monitoring and periodic model updates to maintain performance quality as the nature of event feedback language evolves.

7.6 Visualization of Results

The system provides comprehensive data visualizations through the organizer analytics dashboard. The registration trend chart displays cumulative and daily registration counts for each event, enabling organizers to identify peak registration periods and adjust event promotion strategies accordingly. The attendance visualization presents a breakdown of registered versus attended



participants, providing an attendance conversion metric that helps organizers assess the effectiveness of their registration process. The sentiment distribution chart displays the proportion of positive, neutral, and negative feedback responses for each event in an interactive pie chart format, with drill-down capability to view individual feedback entries by sentiment category. Recommendation performance metrics are visualized as click-through rate trends over time, providing insight into the effectiveness of the recommendation engine in driving participant engagement.

7.7 Summary

This chapter presented the results of the experimental evaluation of the proposed system and provided a detailed discussion of the findings. The sentiment analysis module achieved an accuracy of approximately 87 percent, the recommendation engine demonstrated strong precision and recall metrics, and the QR attendance system performed with near-perfect reliability. The discussion contextualized these results within the system's design objectives and identified areas for future improvement. The overall findings confirm that the proposed system represents a meaningful and practically effective contribution to the domain of academic event management.



Chapter 8: Conclusion and Future Work

8.1 Conclusion

The project titled "AI-Powered Smart Event Hosting and Management System" presents a comprehensive and intelligent solution to the persistent challenges of academic event management. By integrating modern web technologies with AI-driven capabilities including NLP-based sentiment analysis and a preference-based recommendation engine, the system delivers a platform that automates the complete event lifecycle and provides organizers and administrators with actionable, data-driven insights.

The proposed system successfully replaces the fragmented collection of forms, spreadsheets, and communication tools currently used in academic institutions with a unified, centralized platform. The QR-based attendance system eliminates manual sign-in processes, reducing errors and enabling real-time attendance monitoring. The automated certificate generation feature removes a significant administrative burden from organizers. The analytics dashboard provides unprecedented visibility into event performance metrics, empowering evidence-based decision-making at both the event and institutional levels.

The experimental evaluation demonstrates that the system achieves strong performance across all AI components, with the sentiment analysis module reaching approximately 87 percent accuracy and the recommendation engine delivering relevant suggestions with a Mean Average Precision of 0.72. These results validate the design choices made in the system's architecture and confirm that the proposed AI techniques are effective in the domain of academic event management.

8.2 Achievement of Objectives

All primary objectives of the project have been successfully achieved. A centralized, web-based event management platform has been designed and implemented, supporting the complete event lifecycle for academic institutions. The QR code generation and verification system automates attendance tracking with near-perfect reliability. The NLP sentiment analysis module classifies



participant feedback into structured sentiment categories, providing organizers with actionable insights. The recommendation engine suggests relevant events to participants based on their participation history. The real-time analytics dashboard delivers live performance metrics to organizers and administrators. Digital certificate generation is automated upon confirmed attendance, eliminating manual distribution workflows.

8.3 Key Results and Outcomes

The system demonstrates effective performance across all evaluated dimensions. The sentiment analysis module achieved an overall accuracy of approximately 87 percent and a macro-averaged F1-score of 0.855, confirming reliable classification across all sentiment categories. The recommendation engine achieved a Precision at 5 of 0.74 and a Mean Average Precision of 0.72, indicating strong relevance of personalized event suggestions. The QR attendance system achieved a scan success rate of 99.2 percent with an average verification time of 1.8 seconds, substantially outperforming manual attendance methods in both speed and accuracy. The integrated platform was successfully deployed on cloud infrastructure, demonstrating the system's scalability and practical deployability.

8.4 Benefits of the System

The proposed system offers significant benefits to all stakeholder groups within academic institutions. Event organizers benefit from reduced administrative workload, real-time event monitoring, and structured feedback insights that support continuous improvement. Participants benefit from personalized event discovery, seamless registration, contactless attendance verification, and prompt digital certificate receipt. Institutional administrators benefit from comprehensive analytics reports that support data-driven event planning and resource allocation. The system collectively contributes to the digital transformation of academic administration and aligns with sustainability goals by reducing reliance on paper-based processes.

8.5 Limitations of the System



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Despite its strong performance and comprehensive feature set, the proposed system has several limitations that should be acknowledged. The sentiment analysis model was trained on a relatively small domain-specific dataset of 2000 entries, and its performance on feedback from highly specialized technical events may require additional fine-tuning. The recommendation engine relies on simulated interaction data for evaluation, and its real-world performance will depend on the quality and volume of actual user participation data collected after deployment. The system requires stable internet connectivity for all operations, limiting its applicability in environments with unreliable network infrastructure. Data privacy compliance requirements may impose additional constraints on data collection and retention policies in certain institutional contexts.

8.6 Future Improvements

The proposed system can be enhanced in several directions to expand its capabilities and improve its performance. Integration with institutional ERP and student information systems would enable automatic population of user profiles and synchronization of academic schedules with event planning workflows. Development of a dedicated mobile application for iOS and Android platforms would improve accessibility for participants, particularly for QR code scanning and real-time event updates. The sentiment analysis module could be enhanced by fine-tuning on larger domain-specific datasets and incorporating multi-lingual support for institutions in non-English speaking environments.

The recommendation engine could be upgraded to incorporate deep learning-based collaborative filtering approaches, such as neural matrix factorization, to improve recommendation quality as user interaction data accumulates. Gamification features including event attendance streaks, achievement badges, and leaderboards could be introduced to further increase participant engagement. Advanced predictive analytics capabilities, such as forecasting expected attendance for planned events based on historical patterns, would support more accurate resource planning. Finally, enhanced data privacy mechanisms including differential privacy and federated learning approaches could be explored to strengthen the system's compliance with evolving data protection standards.



8.7 Summary

This chapter has summarized the key contributions, results, and outcomes of the AI-Powered Smart Event Hosting and Management System project. The system successfully demonstrates that combining modern web technologies with AI-driven capabilities can significantly improve the efficiency, intelligence, and user experience of academic event management. The identified limitations provide clear directions for future research and development, and the proposed enhancements offer a roadmap for evolving the system into an even more capable platform for smart campus administration. This project makes a meaningful contribution to the growing field of intelligent educational technology and establishes a strong foundation for further innovation in AI-powered institutional management systems.



References

- [1] R. Sharma, P. Kumar, and A. Singh, "Smart Event Management Using IoT and Cloud Computing," *International Journal of Computer Applications*, vol. 185, no. 6, pp. 1–8, 2024.
- [2] M. Gupta and S. Verma, "Sentiment Analysis of Student Feedback Using NLP Techniques," *Proceedings of the International Conference on Computational Intelligence and Data Science*, 2023, pp. 45–52.
- [3] A. Patil and K. Rao, "A QR Code-Based Attendance System for Academic Institutions," *Journal of Educational Technology Systems*, vol. 51, no. 3, pp. 289–304, 2023.
- [4] L. Zhang, W. Chen, and Y. Liu, "Recommendation Systems in Educational Environments: A Survey," *IEEE Transactions on Learning Technologies*, vol. 17, no. 1, pp. 12–28, 2024.
- [5] P. Kumar and A. Singh, "Event-Driven Architecture for Campus Management Systems," *Proceedings of the International Conference on Software Engineering and Systems*, 2023, pp. 112–119.
- [6] W. Chen and Y. Liu, "Deep Learning for Feedback Classification in Higher Education," *Expert Systems with Applications*, vol. 238, pp. 121920, 2024.
- [7] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.
- [8] F. Ricci, L. Rokach, and B. Shapira, *Introduction to Recommender Systems Handbook*. New York, USA: Springer, 2011.



Publication

Appendix A: List of Publications / Certificates

This section includes details of publications, certifications, and achievements obtained during the course of the mini project. These documents serve as supporting evidence of the work carried out and skills developed throughout the project duration.

A.1 Publications

At the current stage, no formal research paper based on this project has been published in journals or conferences. However, the work carried out in this project has strong potential for future publication in reputed conferences and journals related to AI, web systems, NLP, and educational technology.

The project can be further extended and documented in the form of a research paper by incorporating real-world deployment data, extended experimental results, and comparative analysis with existing systems. This would contribute to academic research in the domain of intelligent event management and smart campus technology.

- Title of Paper: AI-Powered Smart Event Hosting and Management System
- Authors: Ashish Vishwakarma, Chetan Shende, Achal Pandey, Sumit Thakur

A.2 Certifications

The following certifications and learning achievements support the technical knowledge applied in this project:

- Certification in Machine Learning and AI (Coursera / NPTEL)
- Certification in Full Stack Web Development – React.js and Node.js (Udemy)
- Certification in Cloud Computing Fundamentals (AWS Academy)
- Certification in Natural Language Processing (DeepLearning.AI / Coursera)



A.3 Project Contributions

During the development of this project, the following contributions were made by the team:

- Design and implementation of the full-stack web-based event management platform
- Integration of NLP-based sentiment analysis using fine-tuned BERT models
- Development of the QR code generation and verification attendance system
- Implementation of the preference-based event recommendation engine
- Design and deployment of the real-time analytics dashboard
- Cloud infrastructure setup and containerized deployment using Docker

A.4 Summary

This appendix provides supporting information related to publications and certifications associated with the project. It highlights the academic and technical efforts of the team and demonstrates the scope for future research and development in the field of AI-powered academic administration systems.